

Year 7 · Chapter 12 · Ratios, rates and time: answers (teacher copy)

12.01 Ratios (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 3 : 5
2. Ex 1b: 4 : 6
3. Ex 2a: 38 : 17
4. Ex 2b: 17 : 55
5. Ex 3a: 2 : 1
6. Ex 3b: No: the order matters

Easy · Set A

1. 1 : 5
2. 5 : 3
3. 4 : 5
4. 7 : 5
5. 2 : 4
6. 3 : 7

Easy · Set B

1. 13 : 12
2. 12 : 13
3. 13 : 25
4. 12 : 25
5. 1 : 25
6. 25 : 1

Medium · Set C

1. 7 : 4
2. 4 : 11
3. 9 : 5
4. 5 : 16
5. 3 : 2
6. 4 : 9

Challenging · Set D

1. $\frac{1}{5}$
2. 8
3. 3 shaded, 2 unshaded in each 5 (6 : 4)
4. yes, e.g. 2 : 3 as $\frac{2}{3}$ of the second amount

Skill drill 1 (Easy basics)

- | | | |
|------------|-------------|-----------|
| 1. 2 : 2 | 15. 12 : 23 | 29. 2 : 7 |
| 2. 1 : 7 | 16. 14 : 15 | 30. 5 : 1 |
| 3. 4 : 15 | 17. 7 : 16 | 31. 1 : 1 |
| 4. 5 : 10 | 18. 14 : 1 | 32. 8 : 9 |
| 5. 5 : 5 | 19. 10 : 12 | 33. 8 : 1 |
| 6. 8 : 10 | 20. 15 : 26 | 34. 2 : 5 |
| 7. 3 : 15 | 21. 1 : 7 | 35. 5 : 4 |
| 8. 15 : 1 | 22. 15 : 22 | 36. 1 : 9 |
| 9. 7 : 15 | 23. 15 : 11 | 37. 1 : 3 |
| 10. 3 : 8 | 24. 14 : 11 | 38. 6 : 7 |
| 11. 5 : 18 | 25. 7 : 6 | 39. 1 : 1 |
| 12. 5 : 5 | 26. 4 : 1 | 40. 6 : 5 |
| 13. 10 : 2 | 27. 2 : 3 | |
| 14. 7 : 3 | 28. 4 : 9 | |

Skill drill 2 (Medium basics)

- | | | |
|------------|---------------|---------------|
| 1. 2 : 5 | 9. 9 : 4 | 17. 1 : 1 : 3 |
| 2. 2 : 3 | 10. 1 : 30 | 18. 4 : 2 : 5 |
| 3. 1 : 3 | 11. 1 : 3 | 19. 2 : 2 : 3 |
| 4. 1 : 6 | 12. 1 : 4 | 20. 5 : 5 : 3 |
| 5. 7 : 6 | 13. 7 : 100 | 21. 1 : 1 : 2 |
| 6. 1 : 200 | 14. 14 : 3 | 22. 3 : 2 : 1 |
| 7. 2 : 15 | 15. 9 : 500 | 23. 6 : 1 : 6 |
| 8. 1 : 3 | 16. 5 : 1 : 6 | 24. 1 : 3 : 2 |

25. 1 : 4 : 4

26. 5 : 2 : 1

27. 5 : 4 : 4

Extension

1. $\frac{3}{10}$
2. 30%

Exit ticket

1. Level 1: 6 : 5
2. Level 2: 5 : 11
3. Level 3: 6 : 7

12.02 Equivalent and simplified ratios

(booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: 8 : 12
2. Ex 1b: 6 : 9 : 12
3. Ex 2a: 4 : 7
4. Ex 2b: 3 : 1 : 4
5. Ex 3a: 2 : 1
6. Ex 3b: 1 : 3

Easy · Set A (quick drill)

- | | | |
|-------|----------|---------------|
| 1. 12 | 5. 9, 12 | 9. 1 : 4 |
| 2. 10 | 6. 3 | 10. 3 : 2 |
| 3. 15 | 7. 2 : 3 | 11. 2 : 3 : 5 |
| 4. 6 | 8. 2 : 3 | 12. 3 : 4 |

Easy · Set B

1. 4 : 7
2. 1 : 3
3. 1 : 6
4. 5 : 9
5. 3 : 1 : 4
6. 1 : 5 : 20

Medium · Set C

1. 2 : 1
2. 10 : 1
3. 5 : 1
4. 1 : 20
5. 1 : 5
6. 1 : 2

Challenging · Set D

1. 1 : 3
2. 4 : 7
3. 48 : 5
4. 3 : 5

Skill drill 1 (Easy basics)

- | | | |
|------------|-------------|--------|
| 1. 7 : 8 | 15. 2 : 1 | 29. 14 |
| 2. 2 : 9 | 16. 2 : 9 | 30. 63 |
| 3. 11 : 3 | 17. 1 : 11 | 31. 3 |
| 4. 3 : 7 | 18. 11 : 12 | 32. 32 |
| 5. 5 : 4 | 19. 8 : 1 | 33. 28 |
| 6. 11 : 7 | 20. 8 : 5 | 34. 16 |
| 7. 7 : 5 | 21. 1 : 7 | 35. 2 |
| 8. 12 : 11 | 22. 3 : 8 | 36. 24 |
| 9. 1 : 5 | 23. 3 : 5 | 37. 24 |
| 10. 1 : 11 | 24. 1 : 4 | 38. 7 |
| 11. 5 : 9 | 25. 8 | 39. 8 |
| 12. 11 : 1 | 26. 18 | 40. 18 |
| 13. 6 : 7 | 27. 64 | |
| 14. 5 : 12 | 28. 48 | |

Skill drill 2 (Medium basics)

- | | | |
|-----------|---------------|---------------|
| 1. 3 : 7 | 12. 4 : 1 | 23. 5 : 6 : 3 |
| 2. 9 : 4 | 13. 2 : 1 | 24. 4 : 1 : 1 |
| 3. 5 : 1 | 14. 2 : 1 | 25. 3 : 6 : 2 |
| 4. 5 : 6 | 15. 7 : 5 | 26. 7 : 6 : 1 |
| 5. 2 : 5 | 16. 6 : 1 | 27. 1 : 6 : 6 |
| 6. 2 : 1 | 17. 1 : 1 : 4 | 28. 6 : 2 : 3 |
| 7. 3 : 2 | 18. 1 : 2 : 1 | 29. 6 : 7 : 5 |
| 8. 8 : 3 | 19. 3 : 2 : 4 | 30. 5 : 5 : 3 |
| 9. 9 : 8 | 20. 1 : 7 : 1 | 31. 1 : 5 : 4 |
| 10. 5 : 8 | 21. 6 : 1 : 1 | |
| 11. 9 : 2 | 22. 1 : 5 : 6 | |

Extension

- 6 : 5
- 1 : 3 : 6

Exit ticket

- Level 1: 21
- Level 2: 2 : 5
- Level 3: 3 : 20

12.03 Ratio problems (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: 100
- Ex 1b: \$45 000
- Ex 2a: \$20 and \$40
- Ex 2b: \$200 and \$250
- Ex 3a: 1.5 km
- Ex 3b: 12 kg

Easy · Set A (quick drill)

- | | | |
|-------|---------------|----------------|
| 1. 16 | 5. 20 | 9. \$15, \$45 |
| 2. 12 | 6. 45 | 10. \$24, \$36 |
| 3. 10 | 7. \$30, \$30 | 11. \$10, \$50 |
| 4. 48 | 8. \$20, \$40 | 12. \$35, \$25 |

Easy · Set B

- \$20, \$40
- \$40, \$60
- \$20, \$25
- 45, 27
- \$200, \$800
- 8, 16, 24 kg

Medium · Set C

- 100
- \$45 000
- 15 L
- \$200, \$250
- 1.5 km
- 12 kg

Challenging · Set D

- 320
- Sam \$36, Jo \$60
- 750 g
- 30 cm by 45 cm

Skill drill 1 (Easy basics)

- | | | |
|------------------|-------------------|-------------------|
| 1. \$20 and \$15 | 9. \$24 and \$32 | 17. \$40 and \$30 |
| 2. \$44 and \$11 | 10. \$16 and \$8 | 18. \$45 and \$45 |
| 3. \$54 and \$36 | 11. \$80 and \$40 | 19. \$30 and \$15 |
| 4. \$3 and \$9 | 12. \$15 and \$45 | 20. \$18 and \$9 |
| 5. \$15 and \$75 | 13. \$36 and \$18 | 21. \$18 and \$18 |
| 6. \$16 and \$4 | 14. \$4 and \$8 | 22. \$54 and \$72 |
| 7. \$33 and \$11 | 15. \$10 and \$4 | 23. \$2 and \$4 |
| 8. \$6 and \$24 | 16. \$8 and \$24 | 24. \$14 and \$7 |

- | | | |
|--------|--------|--------|
| 25. 28 | 29. 32 | 33. 45 |
| 26. 8 | 30. 18 | 34. 21 |
| 27. 27 | 31. 32 | 35. 12 |
| 28. 16 | 32. 6 | 36. 7 |

Skill drill 2 (Medium basics)

- | | | |
|------------------------|-------------------------|-------------------|
| 1. 24 kg, 18 kg, 18 kg | 10. 36 kg, 18 kg, 9 kg | 17. \$6 and \$12 |
| 2. 6 L, 4 L, 2 L | 11. 28 kg, 14 kg, 21 kg | 18. \$48 and \$60 |
| 3. 14 L, 28 L, 21 L | 12. 20 m, 20 m, 10 m | 19. \$20 and \$25 |
| 4. 6 kg, 6 kg, 3 kg | 13. 8 m, 40 m, 24 m | 20. \$6 and \$12 |
| 5. 32 L, 40 L, 8 L | 14. 36 m, 60 m, 24 m | 21. \$20 and \$60 |
| 6. 36 kg, 24 kg, 12 kg | 15. 10 kg, 15 kg, 10 kg | 22. \$52 and \$65 |
| 7. 7 m, 28 m, 14 m | 16. \$44 and \$77 | 23. \$7 and \$14 |
| 8. 32 L, 24 L, 8 L | | 24. \$39 and \$78 |
| 9. 11 m, 44 m, 22 m | | 25. \$48 and \$96 |

Extension

- 12
- A \$28, B \$14, C \$42

Exit ticket

- Level 1: 35
- Level 2: \$30 and \$50
- Level 3: 1 L

12.04 Rates (booklet p. 21)

WE DO examples (teacher)

- Ex 1a: 3000 voters/politician
- Ex 1b: 200 sheep/ha
- Ex 2a: \$0.80/ticket (or 1.25 tickets/\$)
- Ex 2b: 270 stitches/ball
- Ex 3a: 80 km/h
- Ex 3b: 1500 m/min

Easy · Set A (quick drill)

- | | | |
|---------------------------|----------------------|------------------|
| 1. 3000 voters/politician | 4. 270 stitches/ball | 8. 60 km/h |
| 2. 200 sheep/ha | 5. \$15/h | 9. 5 L/min |
| 3. \$0.80/ticket | 6. 80 km/h | 10. \$12/h |
| | 7. \$5/kg | 11. 60 words/min |
| | | 12. \$0.80/pen |

Easy · Set B

- \$5/kg
- 15 L/min
- 90 words/min
- 1.2 beats/s
- \$0.90/L
- 14 km/L

Medium · Set C

- 80 km/h
- 5 L/min
- \$22/h
- 800 km/h
- 25 pages/min
- 5 m/s

Challenging · Set D

- 1500 m/min
- 18 km/h
- 30 c/min
- 20 m/s (= 72 km/h)

Skill drill 1 (Easy basics)

- | | | |
|------------|-----------------|----------------|
| 1. \$55/kg | 2. 18 words/min | 3. 3 words/min |
|------------|-----------------|----------------|

- | | | |
|------------------|------------------|---------------|
| 4. 26 beats/min | 15. 28 beats/min | 26. m/s |
| 5. 7 L/min | 16. 18 beats/min | 27. \$/h |
| 6. 51 words/min | 17. 8 beats/min | 28. cm/week |
| 7. \$12/kg | 18. 57 km/h | 29. c/L |
| 8. 6 km/h | 19. 35 km/h | 30. GB/month |
| 9. 19 m/s | 20. 28 words/min | 31. L/min |
| 10. 54 km/h | 21. \$13/kg | 32. \$/kg |
| 11. 38 L/min | 22. 6 km/h | 33. mm/day |
| 12. 60 beats/min | 23. 36 words/min | 34. L/100 km |
| 13. 40 km/h | 24. \$56/kg | 35. beats/min |
| 14. 40 m/s | 25. words/min | 36. km/h |

Skill drill 2 (Medium basics)

- | | | |
|----------------|--------------------------|---------------------------|
| 1. 0.1 km/min | 13. B (47 and 98.5 km/h) | 19. A (59 and 37.4 km/h) |
| 2. 2 km/min | | |
| 3. 7200 km/h | 14. A (44 and 32.6 km/h) | 20. A (69 and 63.8 km/h) |
| 4. 1800 km/h | | |
| 5. 3.6 c/g | 15. B (36 and 63.3 km/h) | 21. B (71 and 118.7 km/h) |
| 6. 0.3 km/min | | |
| 7. 6 c/g | 16. A (53 and 36.3 km/h) | 22. B (62 and 114 km/h) |
| 8. 5400 km/h | | |
| 9. 0.6 km/min | 17. A (33 and 29.8 km/h) | 23. B (73 and 147.5 km/h) |
| 10. 3600 km/h | | |
| 11. 0.6 c/g | 18. B (72 and 158 km/h) | 24. B (58 and 85.5 km/h) |
| 12. 1.5 km/min | | |

Extension

- 54 km/h
- 10.8 m/h

Exit ticket

- Level 1: \$9/kg
- Level 2: 75 km/h
- Level 3: 1000 m/min

12.05 Best buys (booklet p. 27)

WE DO examples (teacher)

- Ex 1a: \$1.30
- Ex 1b: \$1.11; Chow
- Ex 2a: Dairy Farm (\$2.85/L vs \$3.12/L)
- Ex 2b: 3 L (\$2.20/L vs \$2.30/L)
- Ex 3a: e.g. it may go off or they cannot store it
- Ex 3b: 5 for \$6.50 (\$1.30 vs \$1.40)

Easy · Set A (quick drill)

- | | | |
|--------------|--------------|--------------|
| 1. \$0.60 | 5. \$0.75 | 9. \$2.40/L |
| 2. \$0.65 | 6. \$1.40/m | 10. \$0.45 |
| 3. \$3.50/kg | 7. \$1.50 | 11. \$2.50/m |
| 4. \$1.70/L | 8. \$4.50/kg | 12. \$1.60 |

Easy · Set B

- \$1.30
- \$1.11
- \$1.20
- \$0.85
- \$0.80
- \$0.95

Medium · Set C

- Chow
- 1 L (\$2.85/L vs \$3.12/L)
- 3 L
- 800 g (\$0.70 vs \$0.75 per 100 g)
- 10 rolls (\$0.75 vs \$0.80)
- 1.25 L (\$3.60/L vs \$4/L)

Challenging · Set D

- 5 kg (\$2.10/kg)

- 5 c per 100 g
- no: waste, storage, use-by dates
- own answer

Skill drill 1 (Easy basics)

- | | | |
|------------|------------|-------------------|
| 1. \$3.74 | 13. \$7.77 | 25. 4 for \$15.56 |
| 2. \$8.75 | 14. \$5.14 | 26. 5 for \$5.10 |
| 3. \$2.23 | 15. \$1.31 | 27. 2 for \$4.68 |
| 4. \$7.06 | 16. \$3.34 | 28. 2 for \$5.70 |
| 5. \$7.04 | 17. \$2.95 | 29. 5 for \$12.75 |
| 6. \$1.11 | 18. \$6.47 | 30. 6 for \$6.84 |
| 7. \$6.34 | 19. \$8.04 | 31. 3 for \$9.06 |
| 8. \$3.81 | 20. \$8.81 | 32. 4 for \$4.08 |
| 9. \$8.63 | 21. \$0.92 | 33. 2 for \$2.60 |
| 10. \$6.21 | 22. \$5.28 | 34. 6 for \$8.94 |
| 11. \$8.56 | 23. \$4.63 | 35. 6 for \$13.92 |
| 12. \$3.78 | 24. \$3.26 | 36. 4 for \$8.04 |

Skill drill 2 (Medium basics)

- | | | |
|--|--|-------------|
| 1. 750 g (\$1.45 and \$0.76 per 100 g) | 7. 600 g (\$3.78 and \$1.90 per 100 g) | 13. \$2.68 |
| 2. 600 g (\$3.97 and \$1.55 per 100 g) | 8. 1000 g (\$1.22 and \$0.75 per 100 g) | 14. \$6.63 |
| 3. 375 g (\$0.77 and \$0.81 per 100 g) | 9. 750 g (\$1.41 and \$0.47 per 100 g) | 15. \$15.36 |
| 4. 250 g (\$0.67 and \$1.88 per 100 g) | 10. 1000 g (\$0.84 and \$0.37 per 100 g) | 16. \$6.80 |
| 5. 600 g (\$1.27 and \$1.19 per 100 g) | 11. 150 g (\$1.43 and \$1.44 per 100 g) | 17. \$46.38 |
| 6. 250 g (\$1.20 and \$1.51 per 100 g) | 12. 600 g (\$1.29 and \$1.24 per 100 g) | 18. \$44.85 |
| | | 19. \$14.30 |
| | | 20. \$60.85 |
| | | 21. \$10.86 |
| | | 22. \$4.31 |
| | | 23. \$9.68 |
| | | 24. \$9.04 |
| | | 25. \$85.20 |
| | | 26. \$24.50 |
| | | 27. \$43.10 |

Extension

- 1 L: \$1.35 vs \$1.50 per 100 mL, 15 c cheaper
- \$3.00

Exit ticket

- Level 1: \$4.20/kg
- Level 2: 2 L (\$2.70/L)
- Level 3: 1 kg (\$0.90/100 g)

12.06 Rate problems (booklet p. 33)

WE DO examples (teacher)

- Ex 1a: \$853.10
- Ex 1b: 36.3 L
- Ex 2a: 7.5 ha
- Ex 2b: 200 km
- Ex 3a: 3 h
- Ex 3b: 17.28 L

Easy · Set A (quick drill)

- | | | |
|-------------|--------------|----------|
| 1. \$853.10 | 5. 720 words | 9. \$14 |
| 2. 200 km | 6. \$71.60 | 10. 60 L |
| 3. \$14 | 7. \$60 | 11. 4 kg |
| 4. 120 L | 8. 180 km | 12. 4 h |

Easy · Set B

- 36.3 L
- 7.5 ha
- 3 h
- 15 min
- 12 h

6. 25 min

Medium · Set C

1. 17.28 L
2. \$417.50
3. 18 km
4. 3.36 kg
5. 103 680
6. 27 L

Challenging · Set D

1. 30 min
2. \$66.60
3. 10 min
4. 100 m in 12.5 s (8 m/s vs 6.67 m/s)

Skill drill 1 (Easy basics)

- | | | |
|-----------|------------|---------|
| 1. \$150 | 15. 840 km | 29. 2 h |
| 2. \$36 | 16. \$32 | 30. 3 h |
| 3. \$209 | 17. \$42 | 31. 3 h |
| 4. 280 km | 18. \$135 | 32. 1 h |
| 5. \$75 | 19. \$170 | 33. 1 h |
| 6. \$112 | 20. \$48 | 34. 3 h |
| 7. \$138 | 21. \$56 | 35. 6 h |
| 8. 945 km | 22. 75 km | 36. 4 h |
| 9. \$80 | 23. \$99 | 37. 2 h |
| 10. \$22 | 24. 220 km | 38. 3 h |
| 11. \$12 | 25. 5 h | 39. 5 h |
| 12. \$144 | 26. 4 h | 40. 6 h |
| 13. \$36 | 27. 2 h | |
| 14. \$88 | 28. 5 h | |

Skill drill 2 (Medium basics)

- | | | |
|------------|------------|------------|
| 1. 30 km | 9. 135 km | 17. \$600 |
| 2. 22.5 km | 10. 120 km | 18. \$390 |
| 3. 90 km | 11. 60 km | 19. \$1200 |
| 4. 50 km | 12. 225 km | 20. \$609 |
| 5. 67.5 km | 13. 90 km | 21. \$288 |
| 6. 40 km | 14. 75 km | 22. \$135 |
| 7. 45 km | 15. 180 km | 23. \$580 |
| 8. 300 km | 16. \$870 | 24. \$567 |

Extension

1. 360 km
2. after 2 hours

Exit ticket

1. Level 1: \$147
2. Level 2: 3 h
3. Level 3: 31.25 L

12.07 Travel graphs (booklet p. 39)

WE DO examples (teacher)

1. Ex 1a: 22 km
2. Ex 1b: 2
3. Ex 2a: 3 p.m.
4. Ex 2b: 1 p.m. to 2 p.m.
5. Ex 3a: 4 km/h
6. Ex 3b: faster

Easy · Set A

1. 22 km
2. 2
3. 3 p.m.
4. 1 p.m. to 2 p.m. (5 km/h)

Easy · Set A2 (quick drill)

1. 30 km/h

2. 4 km/h
3. 100 km/h
4. 10 km/h
5. 0 km/h (stopped)
6. 60 km/h

Easy · Set B

1. 300 km
2. 1 hour
3. 80 km/h
4. 40 km/h

Medium · Set C

1. travel graph

Challenging · Set D

1. stopped (not moving)
2. returning to the start
3. e.g. drove 160 km, stopped for lunch for 1 h, then drove on
4. 60 km/h

Skill drill 1 (Easy basics)

- | | | |
|-------------|--------------|----------------------|
| 1. 15 km/h | 10. 100 km/h | 19. 60 km/h |
| 2. 100 km/h | 11. 40 km/h | 20. 60 km/h |
| 3. 10 km/h | 12. 90 km/h | 21. 10 km/h |
| 4. 45 km/h | 13. 90 km/h | 22. 40 km |
| 5. 80 km/h | 14. 100 km/h | 23. 120 km |
| 6. 40 km/h | 15. 40 km/h | 24. 5 h |
| 7. 45 km/h | 16. 20 km/h | 25. from 2 h to 3 h |
| 8. 60 km/h | 17. 20 km/h | 26. hour 1 (40 km/h) |
| 9. 10 km/h | 18. 90 km/h | 27. 24 km/h |

Skill drill 2 (Medium basics)

- | | | |
|------------|-------------|----------------------|
| 1. 160 km | 8. 4 h | 15. 40 km |
| 2. 4 h | 9. 4 h | 16. 20 km |
| 3. 250 km | 10. 120 km | 17. 110 km |
| 4. 80 km/h | 11. 2 h | 18. 6 h |
| 5. 40 km/h | 12. 3 h | 19. from 1 h to 2 h |
| 6. 20 km/h | 13. 80 km | 20. hour 3 (40 km/h) |
| 7. 150 km | 14. 50 km/h | 21. 18.3 km/h |

Extension

1. graph
2. 10 a.m.

Exit ticket

1. Level 1: stopped
2. Level 2: 15 km/h
3. Level 3: 3 km

12.08 Time (booklet p. 47)

WE DO examples (teacher)

1. Ex 1a: 10 h
2. Ex 1b: 3 h; 12 min
3. Ex 2a: 5 h 16 min
4. Ex 2b: 3 min 28 s
5. Ex 3a: 10 h 5 min
6. Ex 3b: 29 min

Easy · Set A (quick drill)

- | | | |
|---------|---------------|----------------|
| 1. 10 h | 5. 2 h | 9. 180 s |
| 2. 3 h | 6. 12 h | 10. 2 min 30 s |
| 3. 4 h | 7. 120 min | 11. 30 min |
| 4. 8 h | 8. 1 h 30 min | 12. 24 h |

Easy · Set B

1. 5 h 16 min

2. 3 min 28 s
3. 150 min
4. 195 min
5. 24 min
6. 1440 min

Medium · Set C

1. 10 h 5 min
2. 4 h 42 min
3. 2 h 9 min
4. 29 min
5. 7 h
6. 1 h 30 min

Challenging · Set D

1. 9:53 p.m.
2. 10 min 55 s
3. 15 s
4. 2 h 51 min

Skill drill 1 (Easy basics)

- | | | |
|----------------|----------------|----------------|
| 1. 120 min | 13. 3 h | 25. 3 h 2 min |
| 2. 360 s | 14. 480 s | 26. 1 h 36 min |
| 3. 6 h 20 min | 15. 180 min | 27. 1 h 56 min |
| 4. 8 h | 16. 7 h | 28. 2 h 45 min |
| 5. 5 h | 17. 420 s | 29. 4 h 9 min |
| 6. 240 s | 18. 9 h | 30. 2 h 15 min |
| 7. 540 s | 19. 240 min | 31. 2 h 25 min |
| 8. 480 min | 20. 2 h 10 min | 32. 5 h 35 min |
| 9. 4 h 10 min | 21. 6 h | 33. 4 h 1 min |
| 10. 180 s | 22. 2 h | 34. 2 h 53 min |
| 11. 7 h 30 min | 23. 2 h 50 min | 35. 2 h 32 min |
| 12. 300 min | 24. 300 s | 36. 4 h 55 min |

Skill drill 2 (Medium basics)

- | | | |
|----------------|----------------|----------------|
| 1. 31 min | 11. 3 h 2 min | 21. 5.4 h |
| 2. 2 h 58 min | 12. 1 h 54 min | 22. 3.1 h |
| 3. 51 min | 13. 6.4 h | 23. 2.5 h |
| 4. 1 h 47 min | 14. 4 h 12 min | 24. 5.5 h |
| 5. 1 h 36 min | 15. 6 h 45 min | 25. 2.4 h |
| 6. 1 h 43 min | 16. 5 h 30 min | 26. 6.25 h |
| 7. 46 min | 17. 6 h 24 min | 27. 1.5 h |
| 8. 42 min | 18. 3 h 12 min | 28. 4 h 30 min |
| 9. 18 min | 19. 5.2 h | |
| 10. 1 h 10 min | 20. 6 h 30 min | |

Extension

1. age × 8760 (approx.)
2. 11 days 13 h 46 min 40 s

Exit ticket

1. Level 1: 2 h 30 min
2. Level 2: 4 h 15 min
3. Level 3: 2 h 25 min

12.09 12-hour and 24-hour time (booklet p. 53)

WE DO examples (teacher)

1. Ex 1a: 3:50 p.m.
2. Ex 1b: 1:48 a.m.
3. Ex 2a: 12:30 p.m.
4. Ex 2b: 14:10
5. Ex 3a: 00:19
6. Ex 3b: no confusion between a.m. and p.m.

Easy · Set A (quick drill)

- | | | |
|--------------|---------------|---------------|
| 1. 1:48 a.m. | 4. 12:30 p.m. | 7. 2:30 p.m. |
| 2. 6:24 a.m. | 5. 11:05 p.m. | 8. 9:15 a.m. |
| 3. 3:50 p.m. | 6. 12:40 a.m. | 9. 12:20 a.m. |

10. 19:45

11. 11:05

12. 00:10

Easy · Set B

1. 14:10
2. 05:33
3. 00:19
4. 13:50
5. 23:45
6. 12:05

Medium · Set C

1. 3:45 p.m.
2. 35 min
3. the 15:10 bus (arrives 15:31)
4. every 30 minutes

Challenging · Set D

1. 02:15, 2:15 a.m.
2. 7 h 45 min
3. 23:40
4. 7:15 p.m. (19:15)

Skill drill 1 (Easy basics)

- | | | |
|-----------|---------------|----------------|
| 1. 23:05 | 15. 04:50 | 29. 8:45 a.m. |
| 2. 19:45 | 16. 00:20 | 30. 10:30 p.m. |
| 3. 19:15 | 17. 01:05 | 31. 12:10 p.m. |
| 4. 08:05 | 18. 11:20 | 32. 12:35 a.m. |
| 5. 12:50 | 19. 04:25 | 33. 6:50 p.m. |
| 6. 14:10 | 20. 21:55 | 34. 6:30 a.m. |
| 7. 03:50 | 21. 01:15 | 35. 7:20 p.m. |
| 8. 07:25 | 22. 21:15 | 36. 10:50 p.m. |
| 9. 09:00 | 23. 07:30 | 37. 3:45 a.m. |
| 10. 04:35 | 24. 20:25 | 38. 4:15 a.m. |
| 11. 03:00 | 25. 7:15 p.m. | 39. 3:20 a.m. |
| 12. 02:25 | 26. 2:20 p.m. | 40. 8:10 a.m. |
| 13. 14:40 | 27. 5:55 a.m. | |
| 14. 00:50 | 28. 8:50 a.m. | |

Skill drill 2 (Medium basics)

- | | | |
|----------------|----------------|-----------|
| 1. 12:20 a.m. | 12. 11:10 p.m. | 23. 19:50 |
| 2. 11:10 | 13. 00:05 | 24. 09:45 |
| 3. 11:15 a.m. | 14. 11:25 | 25. 21:15 |
| 4. 00:25 | 15. 00:15 | 26. 19:00 |
| 5. 23:55 | 16. 23:00 | 27. 08:50 |
| 6. 00:45 | 17. 08:10 | 28. 01:30 |
| 7. 11:40 a.m. | 18. 14:30 | 29. 02:55 |
| 8. 11:55 | 19. 02:20 | 30. 11:40 |
| 9. 12:30 a.m. | 20. 08:15 | 31. 05:50 |
| 10. 00:30 | 21. 07:40 | |
| 11. 12:10 p.m. | 22. 10:55 | |

Extension

1. 05:35
2. 515 min

Exit ticket

1. Level 1: 5:25 p.m.
2. Level 2: 20:05
3. Level 3: 00:45

12.10 Time differences (booklet p. 59)

WE DO examples (teacher)

1. Ex 1a: 4 h 20 min
2. Ex 1b: 7 h 35 min
3. Ex 2a: 6 h 30 min
4. Ex 2b: 30 min
5. Ex 3a: 19 years 2 months 13 days
6. Ex 3b: 18 days

Easy · Set A (quick drill)

- | | | |
|---------------|---------------|----------------|
| 1. 4 h 20 min | 5. 5 h 35 min | 9. 3 h 45 min |
| 2. 6 h 55 min | 6. 2 h 25 min | 10. 2 h 40 min |
| 3. 6 h 30 min | 7. 3 h 30 min | 11. 3 h |
| 4. 2 h 25 min | 8. 1 h 15 min | 12. 35 min |

Easy · Set B

1. 7 h 35 min
2. 30 min
3. 11 h 55 min
4. 7 h 45 min
5. 7 h 40 min
6. 5 h 45 min

Medium · Set C

1. 18 days
2. 45 days
3. 8 weeks 4 days
4. 19 years 2 months 13 days
5. own answer
6. 1 day 6 h 35 min

Challenging · Set D

1. 4 h 55 min
2. 33 days
3. 8 years 10 months
4. \$12.80 (4 part hours)

Skill drill 1 (Easy basics)

- | | | |
|----------------|----------------|----------------|
| 1. 8 h | 12. 5 h 20 min | 23. 30 min |
| 2. 4 h 45 min | 13. 5 h 15 min | 24. 1 h 5 min |
| 3. 1 h 25 min | 14. 6 h 15 min | 25. 4 h 5 min |
| 4. 6 h 5 min | 15. 4 h | 26. 1 h 15 min |
| 5. 3 h 50 min | 16. 5 h 25 min | 27. 25 min |
| 6. 7 h 45 min | 17. 7 h 5 min | 28. 2 h 20 min |
| 7. 4 h 25 min | 18. 2 h 45 min | 29. 4 h 5 min |
| 8. 4 h 35 min | 19. 1 h 50 min | 30. 35 min |
| 9. 2 h | 20. 5 h 10 min | 31. 2 h 25 min |
| 10. 5 h 25 min | 21. 5 h 10 min | 32. 55 min |
| 11. 3 h 20 min | 22. 2 h | 33. 3 h 10 min |

Skill drill 2 (Medium basics)

- | | | |
|---------------|----------------|---------------|
| 1. 2 h 50 min | 9. 6 h 25 min | 17. 9:55 a.m. |
| 2. 1 h 15 min | 10. 2 h 15 min | 18. 6:10 a.m. |
| 3. 4 h 55 min | 11. 4 h 5 min | 19. 1:05 p.m. |
| 4. 2 h 35 min | 12. 4 h 25 min | 20. 1:40 p.m. |
| 5. 5 h 50 min | 13. 2:00 p.m. | 21. 9:15 a.m. |
| 6. 3 h 30 min | 14. 10:45 a.m. | 22. 2:30 p.m. |
| 7. 4 h 10 min | 15. 1:45 p.m. | 23. 5:40 a.m. |
| 8. 5 h 5 min | 16. 8:40 p.m. | 24. 2:15 p.m. |

Extension

1. 10:00 a.m.
2. 07:00 the next day (Sydney)

Exit ticket

1. Level 1: 2 h 45 min
2. Level 2: 2 h 40 min
3. Level 3: 15 days

12.11 Timetables (booklet p. 65)

WE DO examples (teacher)

1. Ex 1a: 8:33 a.m.
2. Ex 1b: 2 h 16 min
3. Ex 2a: 7:40 a.m.

4. Ex 2b: 1 h 13 min
5. Ex 3a: the 7:40 a.m. train (arrives 9:56 a.m.)
6. Ex 3b: no confusion between a.m. and p.m.

Easy · Set A

1. 8:21 a.m.
2. 2 h 16 min
3. 9:25 a.m.
4. 1 h 23 min

Easy · Set A2 (quick drill)

1. 45 min
2. 1 h 25 min
3. 1 h 35 min
4. 2 h 16 min
5. 35 min
6. 2 h 9 min

Easy · Set B

1. 8:33 a.m.
2. 2 h 16 min
3. 7:40 a.m.
4. 1 h 13 min
5. 22:01
6. 1 h 35 min

Medium · Set C

1. 27 min
2. 16:18, reaching the Beach at 16:45
3. the 15:40 bus (Shops 16:01); the 16:10 bus reaches the Shops at 16:31, too late
4. 4:10, 4:18, 4:31, 4:45 p.m.

Challenging · Set D

1. 06:50 → 07:15, 07:30 → 07:55, 08:10 → 08:35, 08:50 → 09:15, 09:30 → 09:55
2. own timetable

Skill drill 1 (Easy basics)

- | | | |
|-----------|----------------|----------------|
| 1. 08:19 | 9. 2 h 5 min | 17. 2 h 30 min |
| 2. 09:03 | 10. 3 h 20 min | 18. 2 h 40 min |
| 3. 09:42 | 11. 1 h 10 min | 19. 1 h |
| 4. 09:57 | 12. 15 min | 20. 1 h 15 min |
| 5. 42 min | 13. 1 h | 21. 50 min |
| 6. 21 min | 14. 1 h 45 min | 22. 35 min |
| 7. 23 min | 15. 35 min | 23. 1 h 55 min |
| 8. 19 min | 16. 50 min | 24. 2 h 5 min |

Skill drill 2 (Medium basics)

- | | | |
|--------------------------------|----------------|----------------|
| 1. Bus 3 | 8. 2 h 25 min | 15. 2 h 5 min |
| 2. 09:07 (Bus 3) | 9. 1 h 55 min | 16. 1 h 30 min |
| 3. they all take the same time | 10. 2 h 30 min | 17. 2 h 25 min |
| 4. 4 min | 11. 20 min | 18. 1 h |
| 5. 10:18 a.m. | 12. 55 min | 19. 50 min |
| 6. 42 min | 13. 1 h 35 min | 20. 30 min |
| 7. 2 h 15 min | 14. 1 h | 21. 2 h 5 min |

Extension

1. 8:33 a.m.
2. 10:08 a.m.

Exit ticket

1. Level 1: 10:30 a.m.
2. Level 2: 21 min
3. Level 3: the 6:05 a.m. train